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Ab Initio Whole Cell Kinetic Model of *Haemophilus influenzae* FDAARGOS_1560 (hinSAH26)

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Abstract

Haemophilus influenzae is a human pathogen involved a wide range of infections, and is listed in World Health Organization Bacterial Priority Pathogens List 2024, making it an important organism to study. Whole cell kinetic models (KMs) is a computational tool used study the biology of pathogens and host-pathogen response. Yet, there is no whole cell KM of *H. influenzae* to-date. This study aims to construct a whole-cell KM of *H. influenzae* FDAARGOS_1560 using *ab initio* approach by identifying enzymes from its genome. The resulting simulatable model, hinSAH26, consists of 816 metabolites, 452 enzymes, and 699 enzymatic reactions. This provides a foundation for *in silico* studies of pathogenic metabolism, identification of critical enzymes, and potential antimicrobial intervention exploration.

Keywords: Whole-cell model; Kinetic model; Differential equations; AdvanceSyn Toolkit

Introduction

Haemophilus influenzae has been implicated in many respiratory infections [1]. Before the Hib conjugate vaccine emerged, *H. influenzae* type b was the most common cause of bacterial meningitis in children worldwide [2]. *H. influenzae* is fascinating because of its genomic diversity and variation in encapsulation [3]. The presence of capsule has been shown to be important in biofilm formation [4]; leading to prolonged illness, and increased infection resistance, resulting in severe respiratory disease [5]. *H. influenzae* is also in World Health Organization Bacterial Priority Pathogens List 2024 (WHO BPPL) [6], making it an important pathogen to study [7].

Whole cell kinetic models (KMs) have been used to study the biology of pathogens [8] and host-pathogen response [9] as KMs yield predictions for both fluxes and product outputs, creating a more complete picture of metabolic behaviour [10]. However, there is no KM of *H. influenzae* to-date. Hence, this study aims to construct a KM of *H. influenzae* FDAARGOS_1560 using *ab initio* approach by identifying enzymes from its genome, and identifying the corresponding reaction from KEGG [11]. The result is a whole cell KM of *H. influenzae* FDAARGOS_1560, named as hinSAH26 using the nomenclature proposed by Cho and Ling [12], which consists of 816 me-

tabolites, 452 enzymes with corresponding transcriptions and translations, and 699 enzymatic reactions.

Materials and Methods

Identification of Reactome

The genome of *Haemophilus influenzae* FDAARGOS_1560 (NCBI RefSeq assembly GCF_020736045.1; NCBI GenBank Accession NZ_CP085952.1) was used as source to identify enzymatic genes using the process previously described [13-15]. Briefly, each enzymatic gene was identified as a presence of complete Enzyme Commission (EC) number in the GenBank record and mapped into reaction IDs via KEGG Ligand Database for Enzyme Nomenclature [11]. For example, EC 1.1.1.23 (https://www.genome.jp/entry/1.1.1.23) catalyses reactions R01158, R01163, and R03012; where the substrates and products of each rection can be identified.

Model Development

The model was developed using the ordinary differential equation (ODE) formats described in Sim et al. [16]. BioNumbers provide a useful basis for estimating transcriptional output in *Escherichia coli*. With roughly 3000 RNA polymerases per cell (BioNumbers 106199) [17] and about 25% engaged in active elongation (BioNumbers 111676) [18], and given a polymerization speed of 22 nucleotides per second (BioNumbers 104109) [19] and an average nucleotide mass of 339.5 Da, the overall production rate is about 5600 kDa of RNA per second (≈9.3e-18 grams per second). Scaling this to a cellular volume of 7e-16 litres [20] and distributing it across 4225 protein-coding genes (BioNumbers 105443) [21] produces a synthesis rate of around 2.92 micromolar per gene per second. With an average transcript lifetime of 107.56 seconds (BioNumbers 107666) [22] (0.93% loss per second), the mRNA dynamics become: d[mRNA]/dt = 0.00292 - 0.0093[mRNA]. For translation, typical mammalian rates of about 1000 peptides/mRNA/hour give 0.278 peptides per second (BioNumbers 106382) [23], and E. coli protein degradation at 1% per hour (2.78e-6 per second) (BioNumbers 109924) [24] leads to: d[peptide]/dt = 0.278[mRNA] - 0.00000278[peptide]. These equations were embedded into an ODE system [14, 25] using median values of kcat = 13.7 per second and Km = 1 mM [26], formatted under AdvanceSyn Model Specification [27].

Model Simulation

The constructed model was tested for simulatability using AdvanceSyn Toolkit [27]. Initial concentrations of all mRNA and enzymes were set to 0 mM. Initial concentrations of all metabolites were set to 1 mM except the following which were set to 1000 mM: (I) C00001 (Water), (II) C00002 (ATP), (III) C00003 (NAD+), (IV) C00004 (NADH), (V) C00005 (NADPH), (VI) C00006 (NADP+), (VII) C00007 (Oxygen), (VIII) C00011 (Carbon Dioxide), (IX) C00014 (Ammonia), (X) C00025 (L-Glutamate), (XI) C00031 (D-Glucose), (XII) C00037 (Glycine), (XIII) C00041 (L-Alanine), (XIV) C00047 (L-Lysine), (XV) C00049 (L-Aspartate), (XVI) C00064 (L-Glutamine), (XVII) C00065 (L-Serine), (XVIII) C00073 (L-Methionine), (XIX) C00097 (L-Cysteine), (XX) C00148 (L-Proline), (XXI) C00536 (Triphosphate), (XXII) C00704 (Superoxide), (XXIII) C01153 (Orthophosphoric monoester), (XXIV) C02051 (Lipoylprotein), (XXV) C11143 (Dimethyl sulfoxide), (XXVI) C13378 (alpha,beta-Dihydroxyethyl-TPP), (XXVII) C14818 (Fe2+), (XXVIII) C15602 (Quinone), (XXIX) C15603 (Hydroquinone), (XXX) C15810 (Sulfur-carrier protein), (XXXI) C15972 (Enzyme N6-(lipoyl)lysine), (XXXII) C15973 (Enzyme N6-(dihydrolipoyl)lysine), (XXXIII) C16236 ([Protein]-N6-(octanoyl)-L-lysine), (XXXIV) C16237 (Protein N6-(lipoyl)lysine), (XXXV) C16240 ([Lipoyl-carrier protein]-L-lysine), (XXXVI) C16254 ([Dihydrolipoyllysine-residue succinyltransferase] S-succinyldihydrolipoyllysine), (XXXVII) C16255 ([Dihydrolipoyllysine-residue acetyltransferase] S-acetyldihydrolipoyllysine), (XXXVIII) C16832 ([Protein]-N6-[(R)-dihydrolipoyl]-L-lysine), (XXXIX) C17023 (Sulfur donor), (XL) C17322 (tRNA containing 2-thiouridine), (XLI) C17324 (tRNA adenine), (XLII) C18237 (Molybdoenzyme molybdenum cofactor), (XLIII) C19647 (Epoxyqueuosine in tRNA), (XLIV) C19848 (Adenylated molybdopterin), (XLV) C19871 (Guanylyl molybdenum cofactor), (XLVI) C20446 (tRNA 7-aminomethyl-7-carbaguanine), (XLVII) C20858 (Protein N5-methyl-L-glutamine), (XLVIII) C20864 (5'-Triphospho-[mRNA]), (XLIX) C21440 ([Protein]-S-sulfanyl-L-cysteine), (L) C22150 (Reduced [2Fe-2S] ferredoxin), (LI) C22151 (Oxidized [2Fe-2S] ferredoxin), (LII) C22154 ([Fe-S] cluster scaffold protein carrying a [4Fe-4S]2+ cluster), (LIII) C22155 ([Fe-S] cluster scaffold protein), (LIV) C22157 (Glycine cleavage system H), (LV) C22158 (Lipoyl-carrier protein E2), (LVI) C22159 ([Glycine cleavage system H]-N6-octanoyl-L-lysine), (LVII) C22160

([Lipoyl-carrier protein E2]-N6-octanoyl-L-lysine), (LVIII) C22881 (Iron(III) oxide-hydroxide), (LIX) G11113 (ADP). The model was simulated using the fourth-order Runge-Kutta method [28, 29] from time zero to 3600 seconds with timestep of 0.1 second, and the concentrations of metabolites were bounded between 0 millimolar and 1000 millimolar. The simulation results were sampled every 2 seconds.

Results and Discussion

The annotated genome of *H. influenzae* FDAARGOS_1560 consists of 1889 genes, including 1756 protein coding sequences. 452 unique EC numbers consisting of 699 enzymatic reactions involving 816 metabolites were identified and developed into a model based on AdvanceSyn Model Specification [27]. In addition, 904 ODEs acting as placeholder for enzyme transcriptions and translations were added.

Simulation of the hinSAH26 model using AdvanceSyn Toolkit [27] proceeded without errors, with Figure 1 illustrating the model's ability to run through its full dynamic course. This validates not only the syntax but also the structural coherence of the underlying kinetic architecture as argued in recent model constructions [13, 15, 30-34]. While the simulations show acetaldehyde plateaus at half the concentration of AMP, such behaviour is likely driven by our decision to use uniform median turnover numbers and Michaelis-Menten constants [26], which erase enzyme-level specificity. Consequently, these flux patterns should not be interpreted as reflecting the organism's physiology. Instead, this model forms a robust baseline whole-cell kinetic framework for *H. influenzae* FDAARGOS_1560, which can be enriched with regulatory networks, growth processes, or experimental kinetic datasets to support more detailed systems-level investigations processes [35-37].

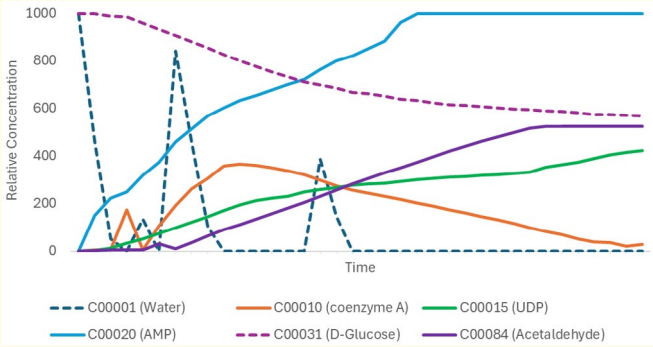

Figure 1: Selection of Simulation Results.

Conclusion

In this study, we present an *ab initio* whole cell kinetic model of *H. influenzae* FDAARGOS_1560, hinSAH26; comprising of 816 metabolites, 452 enzymes with corresponding transcriptions and translations, and 699 enzymatic reactions.

Supplementary Materials

Reaction descriptions and model can be download from https://bit.ly/hinSAH26.

Conflict of Interest

The authors declare no conflict of interest.

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